

ESTABLISHED IN 1861 THE AMERICAN OLDEST BEE PAPER IN AMERICA

BEE JOURNAL

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The Project Gutenberg eBook of The American Bee Journal, Volume XXXIII, No. 4, January 25, 1894

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Title: The American Bee Journal, Volume XXXIII, No. 4, January 25, 1894

Author: Various

Editor: George W. York

Release date: April 15, 2019 [eBook #59284]

Language: English

Other information and formats: www.gutenberg.org/ebooks/59284

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AMERICAN BEE JOURNAL, VOLUME XXXIII, NO. 4, JANUARY 25,
1894 ***

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THE AMERICAN BEE JOURNAL

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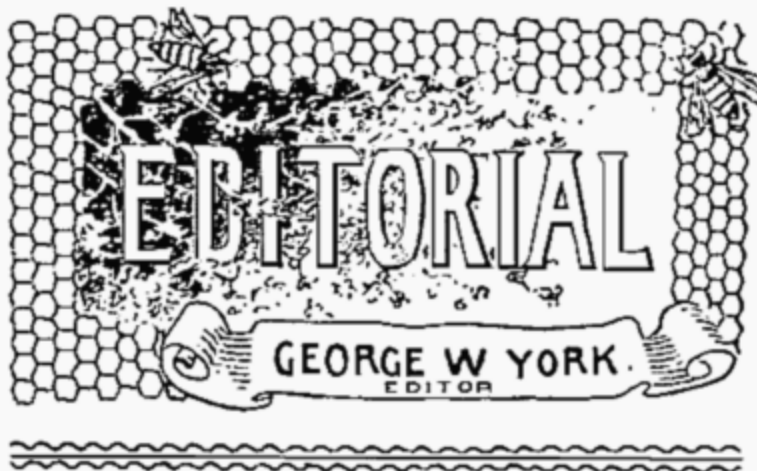
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EDITORIAL

GEORGE W YORK.

EDITOR

Beeswax, so it is said, is formed by one equivalent of starch changed into fat by losing one equivalent of carbonic acid and seven equivalents of oxygen.

The Ohio Convention will be omitted this winter. So we are informed by Miss Dema Bennett, the Secretary of the association. She says that the Executive Committee has so decided, but will hold one next winter. Due notice of time and place will be given in the BEE JOURNAL.

Bro. Geo. W. Brodbeck, of Los Angeles, Calif., has been appointed chairman of a committee to secure and put in place the bee and honey exhibit at the Midwinter Fair now being held in San Francisco.

It is proposed that a "honey pyramid," consisting of comb and extracted honey, be built, six feet square at the base, and 15 feet high. It is thought that 1,500 pounds will be required, and that this will exceed the famous Egyptian pyramids—in sweetness.

California bee-keepers are invited to help make the display, which, no doubt, they will do in a handsome manner.

Bro. G. M. Doolittle is writing a series of semi-political articles for the *Free Press* of Skaneateles, N. Y. The first is on "The Tariff Wrong in Principle." Another will be on "The Tariff for Protection Wrong;" the next on "The Tariff for Revenue Wrong;" then will come two articles on "The Liquor Traffic," which will probably be followed by one on the financial situation of our country. If the reader desires to see all these articles, send 25 cents to the *Free Press* for three months subscription, asking the publisher to begin with Bro. Doolittle's first article, then you will have them all. For ourselves, we can say that we are always interested in what Bro. Doolittle may have to say, whether it be on bee-keeping, or anything else.

☞ Bees never puncture fruit, and unless the skin has been broken by other insects or birds, they never molest it.—*Newman*.

Basis of Honey-Predictions.—Finally, we think we have learned upon what basis certain honey-prophets base their prophecies about honey crops. It appears to be something like this:

The more rain and snow in November and December, the more honey there will be the following season; and if there is no rain or snow in the two months mentioned, there will be no honey.

The predictions are made upon the reports of the State Weather Bureaus, or the Weather Bureau reports in Washington, D. C. All who wish to test the reliability of such a basis for a honey-prediction, should get the weather reports, and begin to foretell for themselves, and thus not be required to await the movements of some so-called "honey-prophet."

We believe the above rule for prophesying is for linden, sourwood, and white clover honey.

Who knows but this may be the secret to which the Tennessee honey-prophet, Sam Wilson, has been so tenaciously hanging on? We shouldn't be a bit surprised if it should prove to be that very secret. If so, every bee-

keeper can now be his own "honey-prophet"—whether he gets any honey or not.

Mr. N. W. McLain—once in charge of a United States experiment apiary, and an apicultural writer—has been visiting recently at Mrs. Atchley's home. Mr. McLain's address is Hinsdale, Ills.

The Iowa Honey Exhibit at the World's Fair, we have pleasure in illustrating and describing this week. No separate appropriation was made for the exhibit, but the Iowa Columbian Commission, recognizing bee-culture as one of the many agricultural pursuits of the State, desired that a creditable exhibit of honey and wax should be made in connection with their agricultural exhibits, and for that purpose appointed Bro. E. Kretchmer, of Red Oak, Iowa, on Jan. 14, 1893. This being too late to secure suitable honey for an exhibit from the crop of 1892, only enough was placed in the case at the beginning of the Fair to retain the space.

There being no money to buy the honey for a suitable exhibit, Mr. Kretchmer, by issuing several circulars, and making several personal visits to prominent apiarists, enlisted the aid of the Iowa bee-keepers, and nobly did they respond by loaning the honey that was exhibited in the Iowa case. Believing that much credit is due those who thus generously loan honey for exhibition purposes, we give the names of those who aided thus, and also what they contributed:

E. J. Cronkleton, of Dunlap—100 pounds of nice comb honey.

F. A. Beals, of Salix—480 pounds of extracted basswood honey, and 544 pounds of comb honey.

R. B. Arnold, of Foster—20 pounds of white clover comb honey.

T. C. DeClercq, of DeSoto—60 pounds of extracted clover honey, 60 pounds of extracted basswood honey, and 105 pounds of comb honey.

L. G. Clute, of Manchester—20 pounds of very nice comb honey, and this was honored with an award.

A. J. Duncan, of Hartford—50 pounds of extracted basswood honey.

F. Furst, of Adair—40 pounds of comb honey.

Oliver Foster, of Mt. Vernon—30 pounds of comb honey, and 60 pounds of extracted honey; which also received an award.

Thos. O. Hines, of Anamosa—91 pounds of comb honey.

Thos. Johnson, of Coon Rapids—22 pounds of comb, and 25 pounds of extracted honey.



Iowa Exhibit at the World's Fair.

Noah Miller, of North English—48 pounds of white clover comb honey.

J. H. Stanford, of Cherokee—20 pounds of aster honey, gathered in October, 1892.

J. L. Strong of Clarinda—100 pounds of comb honey.

J. H. Stephens, of Riverton—60 pounds of extracted basswood honey, and 43 pounds of comb honey.

Mont. Wyrick, of Cascade—100 pounds of extracted honey.

E. Kretchmer, of Red Oak—100 pounds of alfalfa comb honey, which received an award; also 200 pounds of extracted clover honey, which also received an award, and 175 pounds of clover comb honey.

Wm. Kimble, of DeWitt—77 pounds of comb, and 66 pounds of extracted honey, which received an award.

All of the extracted honey was displayed in 18 different vessels, holding from 4 ounces to 12 pounds each.

The principal display of extracted honey was near the east end of the case, arranged on cone-shaped shelving. This cone of honey reached a height of about 6 feet, and was 5 feet in diameter, while near the west end of the case a pyramid was erected, with extracted honey in different sized glass jars, with sheets of glass between the several tiers, and large vase-shaped jars filled with honey were dispersed within the case.

The arrangement of displaying the comb honey consisted principally in a bank near the center of the case, about 11 feet long, tapering from a base nearly 5 feet wide to a crest 6 feet high, surmounted with extracted honey in ornamental jars.

The front, or west end, display consisted of two columns of comb honey which supported the words "IOWA HONEY," built of comb honey; over which was shown a hollow tri-angle of fine comb honey, reaching to the top of the case.

In the east end of the case was displayed a tri-angle of comb honey in open sections, the three walls being 3×5 feet each, surmounted with extracted honey in vessels of various sizes and shapes. Near the edge of the ceiling of the case were suspended neat glass pails filled with extracted honey; while nice specimens of bright wax, in ornamental forms and shapes, were placed in every available nook and corner of the case.

To the untiring efforts of Bro. Kretchmer belongs the credit of securing and placing the very tasty exhibit of Iowa honey and wax. Few men would have undertaken the task, and carried it to as successful a completion, as he did. On another page of this issue of the BEE JOURNAL may be found a picture and also biographical sketch of Bro. Kretchmer—the man to whom Iowa bee-keepers now owe a debt of gratitude.



Fine Weather in Texas was reported by Mrs. Atchley on Jan. 12th. They had had no frost, and everything was green. Cabbage heads weighing 15 pounds were then standing in the gardens. It seems from this that things down there "stand on their heads," and grow just as well as if "right side (or end) up." Good for Texas!





GENERAL QUESTIONS

ANSWERED BY
DR. C. C. MILLER,
MARENGO, ILL.

In this department will be answered those questions needing IMMEDIATE attention, and such as are not of sufficient special interest to require replies from the 20 or more apiarists who help to make "Queries and Replies" so interesting on another page. In the main, it will contain questions and answers upon matters that particularly interest beginners.—Ed.

Feeding Bees in the Cellar.

What is the best way to feed bees when wintering in the cellar? I started in 1891 with three colonies, but have had bad luck the last two winters. I have 13 colonies at the present time—9 outdoors, and 4 in the cellar.

M. W.

Sterling, Ill.

ANSWER.—There is nothing better than to give them frames filled with honey. Carefully remove two or three of the empty combs till you strike the

brood-nest—that is, till you come to a comb with bees on—then put in a comb of honey. Be sure that there are bees enough on the comb next to the honey so that there will be no doubt about their commencing on the honey right away, for there might be such a thing as their starving without ever touching it, unless it were pushed right under their little noses. If the first frame in the brood-nest has too few bees, it may be best to put the honey between this and the next comb with bees on. If the temperature of your cellar varies, don't take the time when it is coldest.



Bees Dying in the Cellar.

As I have always had bad luck in wintering bees out-of-doors, I thought I would winter them in the cellar this winter, so I have followed the *AMERICAN BEE JOURNAL* along, and studied closely all that has appeared in it about wintering bees in the cellar. I finally settled upon the way of one writer's plan, thinking it perhaps a good way; that is, to raise the hive from the bottom-board about an inch, by putting blocks under the corners. I believe he said that the bees would not come out if kept in the dark—that I have done, and I find every time I go to them, that the ground is covered with dead bees, and at this rate I think there will not be any left by spring. Why do my bees come out, if others do not?

My crop of honey last season was none, as usual, and I have had to feed to carry the bees through the winter.

E. H. H.

St. Johnsbury Center, Vt.

ANSWER.—Whatever may be the cause of your bees dying, you may rely on it that it is not likely to be caused by the raising of the hive as you have done. You can keep the bees from troubling the cellar-bottom by letting the hive down and shutting the bees in with wire cloth, for in that case the bees cannot get out, but they'll die just as fast as ever and perhaps a little faster, for if bees find they are fastened in, they are all the more eager to get out.

There must be something wrong with the bees or the cellar, and knowing nothing about it but that the hive is raised and the bees are dying, it is hard to make any kind of a guess as to the cause of the trouble. If the hive is

spotted and daubed about the entrance, then diarrhea is at work. This may come from the quality of the food, or it may come from the condition of the cellar; 45° seems to be the degree of temperature that most agree upon as desirable, and if your cellar is much above 50° it is probably too warm. If below 40°, I would try heating it up, to see what the effect would be. Anything that quiets the bees down may be considered a move in the right direction, for undoubtedly bees that are coming out and dying in large numbers are not quiet.

It may be that the air of the cellar is bad—too close, or poisoned with decaying vegetables—but there isn't much use in going on guessing. The most that can be said is to try to have pure air at about 45°, and see whether the trouble continues.



Increasing an Apiary—Sweet Clover.

1. I have 12 colonies of bees, and I want to increase them. Would I better keep the bees confined to the brood-chamber till they swarm, or would it be best to give them access to one section-case? Or would I better increase by dividing?
2. Where can I get sweet clover seed, that Mr. Newman recommends so highly for bee-pasturage?

J. S.

Long, W. Va.

ANSWERS.—1. If you are anxious for increase and care little for honey, you will do well to put on no supers. Or, you might let part of them have one super each. You may find, however, that it will not make much difference.

If you have had no experience in that line, perhaps it will be best not to meddle much with dividing colonies, but let the bees swarm at their own sweet will; still, it will be good practice for you to make a few colonies by dividing. In any case, be sure to get some good text-book and study up thoroughly, then you will have a more intelligent idea of the whole business. Any points that are not clearly understood will be cheerfully explained in this department, unless you ask too hard questions.

2. Melilot, or sweet clover, seed can be had at any large seed-store. Many of the supply dealers who advertise in the BEE JOURNAL have it for sale.

Carrying Out Dead Brood.

I have a case of dead brood on my hands, which I do not understand. The bees are in a 10-frame hive that I bought of a neighbor last June. At this time they have a hive full of honey and young bees; they are carrying out young brood that is just ready to cap over. It does not appear to be foul brood, and I do not know what to call it. Any information on the subject would be appreciated.

M. F. B.

Indianapolis, Ind.

ANSWER.—It is possible that worms are at work, and the brood is thrown out where they have gnawed away the cappings or some part of the cells. Possibly a cold spell may have made the cluster contract so much as to leave the brood unprotected, when the brood was chilled and afterward thrown out.

Capons and Caponizing, by Edward Warren Sawyer, M. D., Fanny Field, and others. It shows in clear language and illustrations all about caponizing fowls; and thus how to make the most money in poultry-raising. Every poultry-keeper should have it. Price, postpaid, 30 cents; or clubbed with BEE JOURNAL one year, for \$1.10.



BIOGRAPHICAL

Langstroth

No. 66.—Edward Kretchmer.

The subject of our sketch this week is another of the leading bee-keepers whom we had the pleasure of meeting often during the World's Fair last summer, and whom it was always a delight to see on our weekly visits to the apiarian department in the "White City."



E. KRETCHMER.

The *Progressive Bee-Keeper*—the paper from which we take the subjoined sketch—says this of our Iowa friend:

Mr. Kretchmer is one of the pioneers of Western apiculture, a man of whom the bee-keeping fraternity may be proud, and one who has received many honors, both from those of his calling, and the public at large.

As mentioned on another page, it was Bro. Kretchmer who superintended the Iowa honey exhibit at the World's Fair, and his successful efforts are well attested by a glance at the illustration on page 104, and also by the awards secured. Incidentally, we may say that he is one of the largest manufacturers and dealers in bee-keepers' supplies west of the "Mississippi creek."

As we doubtless could add nothing further of interest regarding Bro. Kretchmer, we are glad to give the sketch referred to before, written by his 12-year-old daughter, Valencia, who is not only a member of the Iowa State

Bee-Keepers' Association, but also of the North American. Here is what she writes about "her papa"—and she ought to know him pretty well:

Edward Kretchmer was born on the Atlantic ocean in 1844, on the American merchant vessel, "Louisiana;" and though of German parentage, he is an American by birth. He was brought up in Selicia, Prussia. His father was one of the prominent bee-keepers of his time, and the originator of the first rolls that manufactured "mid-rib," or our earlier form of comb foundation; which, by writers, is frequently confounded with that of Mehring's, whose invention presented the edge of a full thickness of comb, or the beginning of a top-bar. Hence the German translation, "foundation."

Mr. Kretchmer resided about five miles from Dr. Dzierzon, the world-renowned author and apiarist, and from whom, during a season's stay, he received the first lessons in advanced bee-culture. In 1858 he received a colony of Italian bees as a birthday present, and since that time, with the exception of three years, he has been a breeder of Italian bees.

He came to the United States, and to the State of Iowa, about the year 1859, and in the summer of 1860 purchased the first Italian queen that crossed the Mississippi river. In 1861 he entered the army. During his absence, his father sold the original colony to W. H. Furman, of Cedar Rapids, Iowa, the owner of the Langstroth patent for that State.

After his discharge from the army, he again engaged in bee-culture, and while a visitor at the Iowa State Fair, an incident occurred which brought him to public notice, as a well-informed bee-keeper. The incident referred to was this:

An oddly-dressed man, with bees in his hat, was selling little vials of scented water as a "bee-charm," taking dollars right and left, stating that with it bees were rendered peaceable enough to open a hive without being stung. Mr. Kretchmer remarked that he could do that without the drug. The drug vender promptly challenged the youthful-looking German, no doubt expecting him to "back down" from his statement, but on the contrary Mr. K. secured a little smoke, and promptly opened the colony of bees of another exhibitor, quickly found the queen, and exhibited the combs, covered with bees, to a multitude of spectators, without a sting. He was, in consequence thereof, requested by several to communicate his method through some of the journals, and he soon became a noted writer—writing both in the German and English language. In the older files of the AMERICAN BEE JOURNAL, his name may be found to numerous articles. He also issued "Winke Fur Bienen Zuchter"—"Intimation to Bee-Keepers;" "The Amateur Bee-Keepers' Guide," written in 1866, and "The Bee-Keepers' Guide-Book," the latter, a neat volume of 256 pages, issued in 1872.

In 1867 he removed from eastern Iowa to Coburg, Iowa, where he was postmaster for eight years, mayor of the town, and for two terms chairman of the Board of Supervisors for Montgomery county; he declining a re-election, and also the nomination for State Senator.

The demand for better shipping facilities induced him to remove his entire factory to Red Oak in 1890, which is his present residence.

He is a prominent Odd Fellow, and an enthusiastic Mason, being a member of the Blue Lodge, Chapter and Commandery, as well as of the Degree of Rebecca,

and the Eastern Star, both of which he is now the presiding officer.

On the recommendation of the President of the Iowa Bee-Keepers' Association, he was appointed by the Iowa Columbian Commission to take charge of the Iowa honey exhibit at the World's Fair. This appointment he very reluctantly accepted about the middle of last January, 1893, after which time he labored for the success of that exhibit, without the hope of fee or reward, although a very unfavorable honey season made such an undertaking a difficult task. He is now conducting various experiments with new implements in bee-culture.

VALENCIA KRETCHMER.



IN SUNNY SOUTHLAND

CONDUCTED BY
MRS. JENNIE ATCHLEY,
BEEVILLE, TEXAS.

Bee-Keeping and Poultry for Women.

While I was looking over the index of the BEE JOURNAL for 1893, I came across the above subject, and as I have not time to turn back and find what was said on the subject, I wish to add a word more concerning poultry combined with bees for women.

Now, when we take a right view of our surroundings, and look at human nature a moment, we will see that but very few people wish to engage in a

business that will give no rest at all, as it will sooner or later, become a drag, and we become tired, and long for recreation. Well, I for one do not count idleness recreation; but some other light work for rest of mind and body is sought for, and for women (and I will say men, too) I think poultry is first choice to go with bees. I know that it gives me rest and pleasure when tired of working with the bees, to get some grain and call up the chicks, and pet them awhile, and see to their wants—such as good, comfortable nests, roosts, etc.; and this kind of rest proves to me profitable, as well as rest and pleasure.

I know that too much business of any kind is worse than not enough business, and I think we should avoid having too many irons in the fire at one time, but I do believe that to get the best of enjoyment and pleasure, we should have something else to go with bees, whether it pays or not; and I believe that nearly all successful bee-keepers are lovers of flowers and poultry. Am I right?

I have a flock of fine Plymouth Rock chickens, and a bunch of White Holland turkeys, and, oh! how I do enjoy a rest sometimes among my pet chickens and turkeys!

JENNIE ATCHLEY.

Out With a Load of Preachers.

As I have promised to tell more about southwest Texas, I will proceed by saying that Charlie has just returned (Dec. 28th) with a wagon load of preachers that he carried out hunting two weeks ago. I will relate their success.

They killed 20 wild turkeys, 4 wild geese, and a number of ducks, squirrels, armadillo, and a lot of other small game, and one deer. Charles says that of all the lively crowds he ever saw, it is a wagon load of preachers. They made it a rule that the first man that entered camp with a loaded gun should cook a day, and soon they had plenty of cooks.

Charles says that he and one of the preachers went out one day together, and they espied a deer off 50 or 75 yards, and the preacher was to have first shot, then if he missed his aim Charles was to try it. The preacher's gun snapped, and failed to fire, and then Charles fired away and missed the deer. The preacher told him that he had the "buck ague," and so why he missed it. But Charles says that he was laughing at the hard Sunday-school words the preacher was saying because his gun failed to fire, so it turned out that neither of them bagged that deer.

All the preachers got lost, and lay out one night about 15 miles from camp. They had killed a deer, but could not carry it all. A large 12-spike buck would have weighed 200 pounds. Well, they took off the hams, and took turns about carrying it, and they got bewildered, and night overtook them. But about dusk they espied a log-cabin, and went to it, but found no one at home, and from the best they could make out, it was the home of one of the "fence-riders," or where one of the guards lived that looked after the ranch to keep fence cutters and hunters out. The preachers were so nearly famished and tired that they remained all night at the cabin, and no owner came that night, but the preachers found some dried beef and some flour and black molasses, so they were fixed, and soon filled their empty "bread-baskets," as they called them.

They would not sleep on the bed, as they found a six shooter under the pillow, but they laid on the floor, and one kept watch while the others slept, up until midnight, when all fell asleep. The reason they kept watch was, that they feared the owner would come home and take them for robbers, and fire into them without warning. But no owner came, so in the morning they arose early and ate breakfast, and Charles says the preachers say they left some money and a note on the table, to show the owner that they were not robbers, but they longed for a fence-rider to overtake them, as they had gotten in on forbidden ground, and did not know how to get out. As there were 170,000 acres in the pasture, you see they had a hard time of it.

But before night the following day they reached camp O. K., but nearly worn out, and they said they did not expect to find Charles there, as none of them would have staid alone at that camp among the wolves, panthers, wild cats, cougars, bears, etc. But Charles said he made it all right without any trouble, but the coyote wolves kept him plenty of company with their howling. However, the preachers made up their minds that if that 14-year-

old boy had remained at the camp ten miles from anybody all alone, they would each give him a dollar, which they did, though Charles says he thanked them and offered the money back; but no, they said that a boy with all that courage justly deserved the money, and they insisted on his keeping it, which he did.

They then went into conference, and delegated two to go after the rest of that big fat deer, when lo, and behold, *they* got lost, and just barely made it into camp by night, without finding the treasure. They then called together the "court" and discussed the matter of adjournment, as most of them had to get home in time to preach the Christmas sermon at their several churches. Well, they broke camp on the morning of Dec. 22nd, and Charles started home with his wagon load of Baptist ministers.

You will remember that I have told you of our bee-wagon being enclosed with wire-cloth, and resembles a lion's cage. Well, after they got started on their way home, they concluded to play lion awhile, and one of them was a little fellow, and Charles says that the larger preachers tore his clothes nearly all off of him, and when they arrived at the hotel at Beeville, he wrapped himself up in his overcoat and went in, and they had to go out and buy some clothes for him. They wanted him to preach that night in Beeville, but he would not because they had torn his clothes off.

Charlie says that he has been out with lots of crowds, but the preachers were the liveliest set he ever saw. He says they had him promise to haul them out again a year hence, as they were coming if the Lord was willing, and when they could get as brave a boy as he was they were sure of success. Charles says they were very prompt, and paid him \$1.00 per day, besides the premium for his bravery, and he will surely take them out whenever they come.

The preachers killed so many turkeys that they rotted on their hands, and they agreed to knock the first one down that mentioned turkey in a month.

JENNIE ATCHLEY.

How to Draw Brood.

It will be understood that we keep some out yards to draw brood from to keep up the nuclei in the queen-rearing yards. We injured some colonies very much by injudicious drawing of brood. If we do not wish to run the colonies down to nothing, we should mark X on the top-bars of two or three brood-frames, and do not take them when we are drawing brood. I find that two Langstroth frames in the center of the brood-nest will keep the colony up pretty well, but three are better—that leaves about three frames to draw on, when 8-frame hives are used, as the two outside combs seldom have brood, or not as much as the center ones.

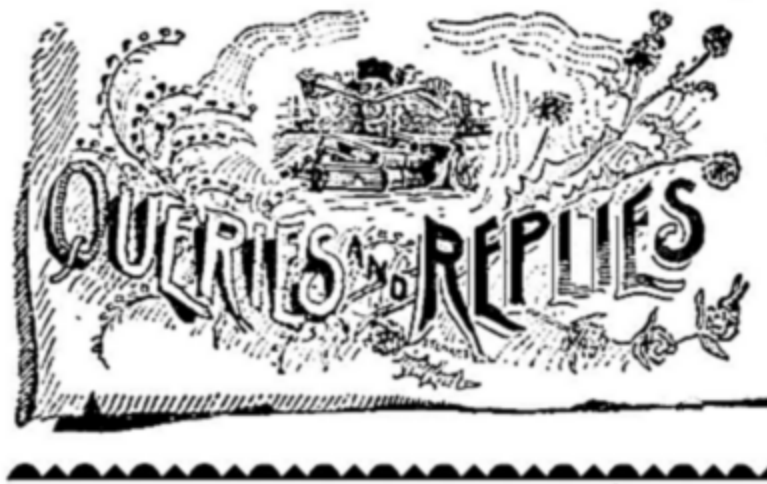
The best plan to control an apiary that persists in swarming, is to draw brood from it and recruit or build nuclei with the brood. It would likely astonish any one to know how much brood can be drawn from a good queen during the season. I am satisfied that we have drawn as much as 50 frames of brood from a single colony during one season of eight to ten months, and then get some honey, and have a fine colony for winter in the colony we draw from. But if we draw at random, and take any and all the frames, we are likely to ruin the colonies.

JENNIE ATCHLEY.



Honey as Food and Medicine is just the thing to help sell honey, as it shows the various ways in which honey may be used as a food and as a medicine. Try 100 copies of it, and see what good "salesmen" they are. See the third page of this number of the BEE JOURNAL for description and prices.





QUERIES AND REPLIES

Color of Queens Regardless of Mating.

Query 907.—If you were buying Italian queens, what color would you expect them to be, regardless of how they are mated?—Iowa.

Yellow.—E. FRANCE.

Yellow, of course.—JAS. A. STONE.

The color of Italian queens.—EUGENE SECOR.

At least three-banded.—J. M. HAMBAUGH.

Any color from yellow to dark leather color.—M. MAHIN.

That depends upon what breeder you are buying from.—P. H. ELWOOD.

I should not anticipate. A leather-colored queen is satisfactory to me.—
MRS. L. HARRISON.

Italians vary very much in color. It is hard to tell in a sentence, what color they should be.—EMERSON T. ABBOTT.

The imported are dark, but American skill has bred them "doubtless pure" to a very bright yellow to the tip.—J. H. LARRABEE.

I should expect them to be somewhere from light yellow to nearly black, and should prefer a dark leather color.—C. C. MILLER.

I should certainly expect them to show three well-developed yellow bands, but would prefer the dark, to the very light yellow queens.—C. H. DIBBERN.

If I "were buying Italian queens," I should *expect* them to be the color of Italians. I prefer such as are known as "leather-colored."—A. B. MASON.

The color is not sure proof, but they should have three distinct yellow bands. They may be brown, light or dark, and still be Italians.—MRS. J. N. HEATER.

Anywhere from a light yellow to a full black, and with all sorts of shades and markings between. Queens may be bred so as to be nearly uniform in color and markings, and the same brood, with a little difference in manipulation, will produce queens several shades darker.—J. A. GREEN.

We do not care for color, if they have the yellow rings, and their workers are gentle and stay on the combs when we raise them out of the hive.—DADANT & SON.

Yellow, or a dark brown color. However, in rare instances I have seen queens as dark as black queens produce fine 3-banded Italian bees.—MRS. JENNIE ATCHLEY.

If I were buying "Italian queens," and knew nothing of their mating, I would expect the three yellow bands, with the other Italian characteristics.—H. D. CUTTING.

I would have to depend on the advertisement of the breeder, and expect what he promised; it might be light or leather-colored, three or five banded.—S. I. FREEBORN.

Anywhere from nearly black to a nearly yellow abdomen, just in accord with their being reared from an imported queen, or the mother of 5-banded bees.—G. M. DOOLITTLE.

Italian queens vary from almost coal-black to almost golden yellow, and in purchasing many I should expect to get almost all shades of color between those two extremes.—R. L. TAYLOR.

I should expect them to show a yellow abdomen all except the tip. But it is said on good authority that some of the imported Italian queens of undoubted purity are quite dark all over.—G. L. TINKER.

Pure Italian queens vary very much in color from bright yellow to dark. Imported queens generally average darker than homebred. I have had some nearly as dark as some black queens.—J. P. H. BROWN.

I should prefer a dark strain, and would expect each worker to be marked with three yellow bands. I have never found the very bright yellow bees so good as gatherers, though usually very amiable.—A. J. COOK.

Of at least three bands of golden yellow; with legs and lower part of the abdomen same color—balance, grayish black. The queen should show nearly the entire abdomen of orange yellow; though the shade varies greatly.—WILL M. BARNUM.

I would expect them to have yellow or leather-colored abdomens, except perhaps some dark color at the tip. *Stripes* around the abdomen is a sign of black blood. But sometimes pure Italian stock will show outcroppings of black blood, and this is often seen in nearly black queens, but such "outcrops" did do it when I reared queens for sale. Remember, the Italian is a "thoroughbred," not a pure-blood race.—G. W. DEMAREE.

I should not care what the color would be. I have found very black queens give very light-colored workers, and *vice versa*. Nothing can be told in this direction from the color of the queen; the mating drone usually governs the color.—J. E. POND.



CONTRIBUTIONS

Selling Extracted Honey at Retail.

Written for the American Bee Journal

BY H. M. MELBEE.

My attention is called to some remarks by Dr. Miller, on page 817 of the BEE JOURNAL for Dec. 28, 1893, on the subject of selling extracted honey. Judging from what I see between the lines, I do not think the Doctor desires any instructions that would insure him 24 cents per pound, at retail, for extracted honey, nor any other price, in fact, whether remunerative or otherwise. The reason, perhaps, for this, is because the Doctor does not produce honey at all in the extracted form, but confines himself entirely to the production of comb honey.

There seems to be all through the Doctor's remarks, a vein of *ridicule*, and this may be accounted for by reason of the fact that certain parties have been, and still are, able to dispose of extracted honey, at retail, at a higher figure than the Doctor can secure for that in the comb. The Doctor says he is sure that he could do nothing of the kind himself, but that should surprise no one, nor should it discourage others from trying to do so. Not many years ago the Doctor stated publicly, if my memory is not at fault, that he could not produce first-grade comb honey under a special classification, whereas there were others who thought they could. And, judging from what was shown at the World's Fair honey exhibit last year, it was plain to see that the Doctor was right in what he thought he could not do in that direction.

Some of the imaginary talk the Doctor gives with that imaginary lady, plainly shows that he has had no experience, worth copying, in selling extracted honey. In short, he imagines a talk, which, in some respects, I have never met with in more than 20 years' experience. An agent, properly instructed, who could not have silenced that lady's remarks, would not be worth his salt to me.

I know from experience, not theory nor imagination, that any lady or gentleman competent to sell books, can be instructed to sell extracted honey in thousands of cities and villages, and at remunerative prices. But the agent must have proper instructions, and then must follow them. But I never attempt to give those instructions to any one who has no desire to know them. I have now had in my employ three agents who have always sold extracted honey at my prices, and profitably, by following my instructions. One of these agents was a lady—the other two young men.

No, Doctor, I do not live in a mountainous country, nor where the people I trade with live miles away from groceries, nor where honey-producers are unknown. The country where I live is just about as level as where the Doctor resides, and groceries are just about as convenient and numerous. I presume the people are just about as wealthy and intelligent, with possibly one or two exceptions, as those in the Doctor's neighborhood. This being the case, the Doctor does not seem to understand why my customers do not find out that they can buy honey at a lower price than they pay me. Why, doctor, they do know they can buy comb honey, in wooden sections, at about the price you mention, but they have intelligence enough to know that when

they pay for a section of wood and honey they do not get, on an average, to exceed 12 ounces of honey. And, with some assistance, they reason thus: If they have to pay 20, or even 18, cents for three-fourths of a pound of honey, they might as well pay my price, or 24 cents, and get 16 ounces, or a full pound.

Again: Neither Melbee nor his agents, when soliciting orders for honey, have ever yet been found guilty of carrying around with them a bee-paper of any description, for the express purpose of showing to would-be purchasers the market reports, as prepared and manipulated by commission merchants, nor do they ever intend to be guilty of doing so. On the other hand, the Doctor perhaps would not approve of such an un-business-like procedure. I presume the Doctor would carry a sample of honey in one hand, and a sample of one of those market reports in the other, and then call the special attention of his patrons to both samples. That, of course, would be just like the Doctor!

The Doctor seems to think that Melbee might be a wealthy man if he would only set a score or so of agents to work selling honey for him on his terms and at his prices. Perhaps the Doctor is right for once. On the other hand, the Doctor perhaps might have been also wealthy, if he had stuck to the music trade at a salary of—say \$2,500 per year. But as he did not do so, perhaps we have a right to infer that he has become exceedingly wealthy from the sale of his honey crops.

The Doctor attempts to make it appear that the difference between 7 cents wholesale, and 24 cents retail, is all profit. A novice might think so, but a bee-keeper of Dr. Miller's experience should know better. Evidently the Doctor has had no experience as to the expense connected with the sale of extracted honey, by the plan pursued by myself and my agents, or else he desires to misrepresent the profits we obtain. I am frank to confess that we do get a good profit, but no larger than thousands of others might secure by knowing how.

To conclude: Melbee desires it to be distinctly understood that he does not follow the honey-trade simply for health and pleasure, but mainly for dollars and cents.

Honeyville, Beeland.

Positive Prevention of After-Swarming.

Written for the American Bee Journal

BY FRANK COVERDALE.

James Heddon, I believe, was the first to give us a practical method for the control of after-swarming; however, the method could not be absolutely depended upon to do the work, but was a grand step in the right direction. Who knows, to a certainty, just when the first queen-cell will hatch in the old hive—whether it will be 5 days or 15 days? A second swarm might issue before the old hive was ever moved to its permanent stand, and again after it had been moved, on account of the first cell hatching so late.

It was when I was busy making hay, when an occasional swarm would leave me, causing much vexation in my mind, and many hours of deep study, how I should overcome this difficulty; and it came, to my mind that a bee-escape might do the work, so I attached one to a hive, at the first opportunity. A 1½-inch hole was bored in the center of one side near the bottom edge, and a wire-screen cone fitted in the hole, and the entrance entirely closed; the newly-hived swarm was placed close by its side, with the entrance just under the above prepared cone, and every bee that left the old hive became an occupant of the new hive.

In three days an examination was made in the old hive, for I was afraid that too many bees would leave the brood, and destruction be the result. But, oh, how I was delighted! All was lovely still. In three days more another examination was made, with like results, and still another three days later, making nine days. Then I began boring holes in other hives, and treating them as above, with the same results, until all (60 colonies) that swarmed were in the same condition.

Some of the old hives were moved to their new stands in 12, others in 13, 14 and 16 days, the last being rather too long a time—14 days is about right in my location. Then these old colonies can be given a ripe queen-cell, or a queen, or the entrance be opened and left so until all the young bees are hatched, when the entrance can be again closed, and it will unite with the new swarm, and the combs will be empty. In fact, you may have full control of the matter, as to managing against second swarms.

The first two years I used the bored holes, covering them over when through, with a piece of section tacked over them; but since then an escape has been used at the entrance; however, at times the latter would get clogged, causing some annoyance, and I now think the bored hole at the side is best.

If the reader will carefully look over the back numbers of the BEE JOURNAL, it will be seen that I have touched upon this point before, but dare not recommend it as being entirely practicable. But I hesitate no longer, but advise all who stand in need, to try and be convinced how this plan lessens labor, cost and vexation of after-swarms; and in my location greatly increases a crop of comb honey, and of finer quality than it otherwise would have been.

Fear not that the new swarms will be overcrowded in numbers, and swarm again, but furnish each new hive with starters below and full sheets of comb foundation in all the sections above, and you will soon begin to wonder whether it is best to "prevent swarming" or not. It is nice to have wood-zinc queen-excluders, then all can be arranged at the time of hiving—such as moving the partly-finished sections from the parent colony immediately to the newly-hived swarm, and not have to wait two or three days for the queen to establish her brood-nest below.

Welton, Iowa.

Making Sugar Syrup for Feeding Bees.

Written for the American Bee Journal

BY G. M. DOOLITTLE.

The following has come to hand from some one who forgot to sign his or her name, so I will answer through the BEE JOURNAL, as requested.

"Will you tell us through the columns of the AMERICAN BEE JOURNAL just how you make sugar syrup for feeding bees, as I have some bees which will need feeding before long? I think you have given this before, but I cannot find where it is. If I remember rightly, you use honey to a certain extent, and, if so, is there no danger of getting foul brood, where one may have to buy honey for this purpose?"

In answering the above, it may be well, and interesting to the reader, to know just how I came to hit on the formula for sugar syrup, which I have given several times before in the different bee-papers, as hinted at by our correspondent.

Some years ago, after a poor season, I found that all of my queen-rearing colonies would have to be fed, as well as some of the others, so I set about looking up recipes for making the feed, as I had no surplus combs of honey. I found plenty of recipes telling how to make it, using vinegar, cream-of-tartar and tartaric acid in greater or less quantities to keep the syrup from candying or crystallizing. When about concluding to use one of these, I ran across one that said all that was necessary to do was to pour boiling water on the granulated sugar, stirring both together as long as the water would dissolve any more sugar. As this seemed so simple I concluded to use this.

Having the syrup made and the feeders in the hive, I proceeded to feed, all going well the first feed. When I came to feed the second night, I found the feed skimmed over with a crust of sugar which had formed on the surface during the 24 hours it had been standing. I also found that it had granulated on the bottom and sides of the can, and upon going to the hives I found a little on the bottom and sides of the feeders. However, I persisted in feeding it, as the one giving the plan said nothing was needed to keep the syrup from crystallizing, as the bees put acid enough into it in manipulating to keep it a liquid.

After a few days, I noticed bees out at the entrance of the hive of each colony fed, having little grains of sugar on their wings and bodies, trying to

fly, but most of them had so much on them that they could only hop around, making a purring sound with their wings. I next looked inside of the hive, when I found that fully one-fifth of the bees had more or less of these sugar crystals on them, while the inside of the feeders was all covered with crystals. Upon looking into the cells containing the syrup, I found that in many of them crystallization had commenced to such an extent that the crystals were easily seen. I said this would not answer, so when the next batch of syrup was made, I put vinegar in the water before stirring in the sugar. While the vinegar helped about the crystals, it also gave a taste to the syrup which I did not like, so in the next I tried cream of tartar, and then tartaric acid; but in spite of them all, the syrup would crystallize some, unless I added so much that a disagreeable taste was given the syrup.

It now came to me, how in early years I had used, owing to scarcity of honey at our house, honey and sugar mixed, on the table, in which case neither the honey nor sugar granulated, so the next batch of syrup was made as follows:

Fifteen pounds of water was put into a large tin dish and brought to a boil, when 30 pounds of granulated sugar was poured in and stirred for a moment till it had mostly dissolved, when it was left over the fire till it boiled again. Upon taking from the fire, five pounds of honey was poured in, and the whole stirred enough to mix thoroughly. I found in this a syrup of about the consistency of honey, which remained a liquid from day to day—a syrup that any bee-keeper could easily make, and one which would not crystallize on the bees, feeders or in the cells. I have kept this syrup standing in an open dish for months at a time without its crystallizing or souring.

It has now been some 10 or 12 years since the experiments above given were tried, and during all that time I have never found how I could improve on this food for feeding bees for winter stores. For spring feeding, I would use 25 pounds of water to the same amount of sugar and honey, as this gives better results in brood-rearing than does the thicker syrup.

As to there being any danger, should it so happen that honey from a foul-broody colony was used, I would say that there need be no fears, for if the honey is stirred in as above given, it will all be scalded, and the scalding of honey anything else having the germs of foul brood about or in it, effectually kills these germs. However, care should be used in handling

honey which may have come from a foul-broody hive, as the least bit of it carelessly left where the bees can get it, while in its raw state, will carry with it the seeds of foul brood, just as surely as corn grows from seed corn.

There is one other item I wish to notice before closing, and that is where our correspondent hints at its being necessary to feed his bees before long. If, as I suspect, the correspondent lives in the North, he should have fed the bees in October what they needed to carry them through the winter. This is a duty he not only owes to himself, but to the bees also, for, while bees often do come through the winter when fed during cold weather, yet the chances are that a loss of colonies will not only waste the bees, but the syrup fed as well.

Borodino, N. Y.

Bees in the Sierra Nevada Mountains.

Written for the American Bee Journal

BY S. L. WATKINS.

The honey season, the past year, was above the average, and bees have paid well. The last, two or three seasons I have not kept so many bees as I formerly did. I took up a piece of land here in the upper Sierras, and have been improving it. I have started in the nursery business, and I find that this goes splendidly with the bee-business. From now on, I shall increase my bees up, and go into the business extensively again. I shall place all my hives in a straight row, and build a car track behind them, and I can easily run all my honey into the extracting house. This will facilitate things greatly. The past summer I had my hives set on stumps, and I tell you it was a job to carry the combs to the house where I extracted.

Some will ask, why I kept my bees on stumps. Well, I had no other place to put them. This country is heavily timbered with yellow pine, sugar pine,

incense cedar, cypress, spruce, fir, madrona, oaks, etc.; and it is extremely hard to clean, but after the land is once cleaned, it is very valuable.

Land that was one year ago covered with pine stumps, is now covered with strawberry plants, fruit trees, and ornamental plants, and they look splendidly, too. It took an immense amount of work, but it pays well.

The bee-hive that I use, and the one that I expect to use for a long time, takes frames about 7x14 inches; the hive is about 14 inches wide; two stories comprise a hive, which is about 16 inches high. I tier up several stories high in the honey season. I find that I can handle bees very rapidly; can shake the bees from the combs without even breaking the comb loose from the frame; with the Langstroth hive, or frame, rather, the combs will give way occasionally in hot weather, if not wired.

I think that I shall always run for extracted honey at this apiary. In Placerville, Calif., where I used to rear bees for sale, I had a decided preference for Carniolan bees. I think that I shall rear them largely. The so-called Golden Italian bees—if I can prove to my satisfaction that they will equal the Carniolans, I will insert a number of queens.

When I was extensively engaged in rearing queen-bees, the call was for the leather-colored Italians—and very few of the light-colored queens were called for; now it seems to be the reverse, all queen-breeders are advertising the Golden Italian bees and queens.

The Holy Land and Cyprian bees seem to have gone out of fashion; so also the Albino. I see that the Carniolan race is not much advertised in the bee-papers any more. Well, I shall pin my faith to them for awhile yet, until I find something better.

A cross between the Carniolan and Italian race of honey-bees, makes wonderfully energetic bees; they protect their hives well, and are marvelous honey-gatherers.

There has been a greater interest manifested in bees this season than for a long time, and I contemplate, from now on, that a great many will engage in this industry in the Sierra Nevada mountains.

I am pleased to see that the AMERICAN BEE JOURNAL is improving so steadily. Later on I shall give you some bee-notes for its columns.

Grizzly Flats, Calif.

Danger in Climbing for Swarms, Etc.

Written for the American Bee Journal

BY LEWIS K. SMITH.

As a caution to my brother bee-keepers, I will say: Don't climb unless you are sure of your footing. On the 26th of last May, I was called on to hive a swarm of bees that had settled high up in an apple tree. Having been a great climber from childhood's days, I bounded up to the top of the tree, and sawed off the limb containing the swarm while I stood on a limb below. The additional weight of the bees and limb split off the one on which I stood, and down came Smith, bees and all—a distance of 22 feet. That it did not kill me I am humbly thankful to Him who holds us in the hollow of his hand. For nearly two days I was partly paralyzed, and was finally relieved by the application of an electric battery, and other means applied by two eminent physicians who were unremitting in their attentions day and night.

Do we sufficiently appreciate the arduous labors of those men of science, working day and night with both mind and body, taxing every energy, and enlisting every sympathy of their being? Is it wonderful that they wear out, break down, and die suddenly? Whenever I think of those terrible hours of suffering, when my digestive system was completely paralyzed, and my life was wavering in the balance, my heart goes out in thanksgiving to a merciful Providence, and I treasure the faithful ministrations of my physicians and friends.

DRUMMING BEES FROM A HIVE WITH CROSSWISE COMBS.

Let me add one suggestion to the instructions to F. M. L., on page 716 of the BEE JOURNAL for Dec. 7, 1893, relative to drumming bees out of a hive with cross combs, into a box above. If he will go to some hive having nice,

straight comb, and get a frame or two with unsealed brood, and put in the upper box or hive, I am of the opinion his bees will go up more readily, and stay better contented. Then, too, it will be much easier to find the queen if he desires to supersede her.

RESULTS OF THE PAST SEASON.

From 66 colonies, spring count, some of them nuclei, my crop of section honey was between 400 and 500 pounds. This surplus was stored by a few extra-good colonies—one of them filling 78 sections, and not swarming. The queen of this colony was reared in 1892 from an imported Italian mother.

Another that did well, was a cross between the Italian and Carniolan stock. On the whole, I find the Carniolans much less desirable than the Italians. It may be that my Carniolans were not pure, but they are extra-good fighters, vindictive, and often pounce on me without provocation. So I've been superseding the queens for two years.

Gainesboro, Tenn.

Brace-Combs—Cause and Prevention.

Written for the American Bee Journal

BY E. J. CRONKLETON.

Brace-combs and bees must be studied in connection, in order to arrive at correct conclusions about brace-combs. I think I will be able to convince every well-informed bee-keeper—one who has well learned the nature, habits and instincts of the bee—that there is some truth in my theory, or at least it will cause him to do a little thinking, and that is healthful.

We all know how irritable the bee is, especially by a jar—the least little jar will bring a response from every bee in the hive. Well, suppose it does, what of that? Well, nothing particular, only we learn something by it. Suppose the

frames are loose at their bearings, resting on metal bearings for your convenience, and a perfect torment to the bees—torment because they cannot glue them down at the ends, and their walking over the combs causes them to tremble, and a bee cannot stand that—it is a constant annoyance. What is the result? Why, brace-combs are the inevitable result? They go right at it, and brace up and strengthen those combs, just as long as there is the least jar or tremble about them.

Years ago, when I was taking my first lessons in the art of bee-culture, I had but few brace-combs; but the metal-bearing craze was sprung on me. I had considerable trouble prying the frames loose from their bearings, so I thought this will be nice—I can just pick the combs out—it will be a pleasure indeed. The result was just as fine a lot of brace-combs as any one ever saw. The bees literally filled the spaces between the top-bars up, only leaving here and there a hole to pass through to the section.

Well, I looked at them as I examined hive after hive, and I thought I was undone entirely. It looked very much as though my elegant scheme had miscarried, while the bees had made a perfect success of theirs, though I, at that time, had not the remotest idea what caused the bees to interlace the combs in that manner.

I could easily see that I would better fall back on first principles, which I did, and brace-combs have disappeared ever since, with me, in proportion to the pains that I have taken to have the combs well fastened in the hives. I have no scheme for fastening the frames, just so they will not tremble and shake when the bees travel over them.

Mr. Heddon's thumb-screw business would be just the thing. The Hoffman frame can be used to advantage. Suit yourselves, and use your own judgment, and your own resources.

I have said nothing about burr-combs, from the fact that I am not certain that I know anything about them. I see a difference, but I think their mission is the same.

Try this, and I am satisfied you will see that I am right for once.

Dunlap, Iowa.

Convention Notices.

WISCONSIN.—The Wisconsin Bee-Keepers' Association will meet in Madison, Wis., on Feb. 7 and 8, 1894. An interesting meeting is expected. It is earnestly hoped there may be a full attendance.

J. W. VANCE, Cor. Sec.

Madison, Wis.

KANSAS.—There will be a meeting of the Southeastern Kansas Bee-Keepers' Association on March 10, 1894, at the apiaries of Thomas Willett, 5 miles northeast of Bronson, Bourbon Co., Kansas. All are invited to come.

J. C. BALCH, Sec.

Bronson, Kans.

A Binder for holding a year's numbers of the BEE JOURNAL we mail for only 50 cents; or clubbed with the JOURNAL for \$1.40.



CONVENTION Proceedings

The Michigan State Convention.

Reported for the "American Bee Journal"

BY W. Z. HUTCHINSON.

The Michigan Bee-Keepers' Association held their 28th annual convention on Jan. 2 and 3, 1894, in the Common Council Chambers in the city of Flint. The convention was called to order by President Taylor, and the following members paid their dues:

M. H. Hunt, Bell Branch.
L. A. Aspinwall, Jackson.
Hon. R. L. Taylor, Lapeer.
Wm. Anderson, Imlay City.
H. D. Cutting, Tecumseh.
W. Z. Hutchinson, Flint.

August Koeppen, Flint.
Earl Post, Atlas.
E. M. Miller, Swartz Creek.
M. S. West, Flint.
H. Webster, Byron.
H. L. Hutchinson, Mayville.
E. G. Grimes, Vernon.
Byron Walker, Evart.
Chas. Koeppen, Flint.
Andre Torry, Flint.
M. McWain, Grand Blanc.
L. H. Root, Prattville.
Jas. Cowe, Imlay City.
Jno. Cowe, Imlay City.

Pres. Taylor then read the following essay, entitled,

Apicultural Work at Experiment Stations.

If I appear to any to go into devious paths in a brief treatment of the topic assigned me, it is owing to the latitude which the topic itself gives me.

And first I ask, do bee-keepers want it? that is, do they want that sort of work at the stations? I am sometimes in doubt about it. I judge somewhat from the course of my own feelings in the matter. Before I became connected with the work and began to study into it, I was not inclined to esteem it over highly, but now if I were to express my thoughts and feelings freely, you would no doubt think me on the verge of the domain whose inhabitants are called cranks. Such is the effect of contact and acquaintance. Now, while the great body of bee-keepers has not the enthusiasm which close contemplation begets, yet if called upon they would vote pretty unanimously in favor of the work.

Then the question suggests itself, why would they vote for it? Provision has been made by the general government by which the agricultural college of each State is to receive annually a certain sum of money to be devoted to the support of an experiment station in the interest of agriculture and kindred pursuits generally. This sum was to be in the first instance, as I

understand it, \$15,000, and after that to be increased by the sum of \$1,000 each year until the amount of \$250,000 is reached, which is then to remain fixed at that point. That is, that is to be the course of affairs, unless the ideas of economy of the present administration at Washington require that this money be kept in the general treasury. This is a considerable sum of money, and apiculture is equitably entitled to all and more than it is now getting in this State.

Now is it simply because they are equitably entitled to it, that the bee-keepers would claim a just share to be devoted to apicultural work, like a school-boy unwilling that his fellow should use his sled whether he wants it himself or not? Or is it because they feel it is not only their right, but to their advantage? Have they such a lively faith in the probable value of results that they will scrutinize and study them? That bee-keepers should have an active interest in these matters is of the utmost importance if the work is to go on. Those in authority are generally quite ready to be directed by the will of those they serve, if they can learn certainly what that will is.

Can the work be made of real value? Take one item. For myself, I have become more and more impressed with the importance of a thorough knowledge of foundations designed for use in sections for the production of comb honey. Much has been guessed, but so far as I can learn little is yet *known* on this subject. In the experiment of which I recently gave an account, one of the objects aimed at was to determine, if possible, if there was a difference among them, and, if so, what kind was of such a nature as to enable the bees to work it down most nearly to the thinness and character of natural comb. To me the results were very satisfactory and encouraging, and this not because one kind was shown to be better than another, but because it appeared that a method had been hit upon by which the relative value of foundations could be practically determined.

But this, it seems, is only a beginning. Now that a door is open, many other questions come up at the very threshold and press for a solution. What makes the difference among foundations? Is it the character of the machine used in making, or the character of the wax? or is it the method of dealing with the wax? Then, if comb from foundation is made as thin as the natural comb, is it still more tenacious, or is it equally friable and tender?

Again, it is well understood that the natural comb is not composed entirely of wax, but that other substances are combined with the wax. Can anything be done to imitate the natural comb in this, and so make foundation even less subject to the charge of being an adulteration than it is at present? This suggests the matter of economy of wax in the use of foundation thus: What is the per cent. of wax wasted, not to say worse than wasted, when so made into foundation that the septa of comb resulting is 60 per cent. thicker than the septa of natural comb? or, to put it in another way, if foundation whose septa the bees will work down to a thinness of 90/10,000 of an inch is worth 60 cents, what is that worth whose septa the bees will work down to a thickness of 60/10,000 of an inch? Probably from 25 to 40 per cent. more. If a man uses much foundation, this should touch him at the tenderest point.

I try not to be carried off my feet by enthusiasm, perhaps, nevertheless, I may be. What do bee-keepers who stand off at arm's length think of the value of such investigation?

It will not do to say it is better not to agitate these and such like questions, it will only call the attention of consumers to the defects of comb honey as now produced, and injure its sale. It can hardly injure the sale of honey for consumers to know that we are trying earnestly to improve its quality, but if on eating it a heavy wad of wax forms in the mouth, that will do the work though the eater may hardly know exactly why. Nothing finds so ready a market as goods that give a fine sensation to the palate in every particular. We are bound to make our comb honey equal in every respect to that produced by the bees unaided by foundation, if we can.

I can think of nothing that would have a greater tendency to popularize the work of the station, and to excite the interest of the bee-keeping fraternity in it, than to enlist as many as possible in the matter of making suggestions as to subjects and methods of experiment, but more especially as to *methods*. Subjects are plentiful and easily discovered, but simple and satisfactory methods are often slow to suggest themselves. I meditated upon the matter all summer before a practical plan for the comparison of combs made from different foundations presented itself; to another mind the first thought would have been the right one.

Now, I am at work endeavoring to discover a method of procedure for determining the cause of the wintering trouble. I want it to be so plain that

every one will recognize it as the right one, and be compelled to accept its utterances as final. It is hardly necessary to say that it is still undiscovered, but perhaps our own journal, the *Review*, might furnish us the key by means of a symposium of numerous brief articles addressed to this one point.

Finally, as a closing paragraph, I want to take this opportunity to make a suggestion to the apicultural journals of the country. I am no journalist—I make no professions of knowing how to conduct a journal, and, I am not going to offer any advice on that point, but I wonder if some of them without detriment to themselves could not give a little more active assistance in sustaining the work by an effort to create a more general interest in its behalf. For that purpose, probably nothing could be better than candid criticism.

R. L. TAYLOR.

At the close of the essay Pres. Taylor remarked: "I suppose it is known that no appropriation has been made to continue this work for more than one year, and whether it is to be continued will depend somewhat upon the action taken by this meeting. It would also be well to have a committee appointed to decide in regard to the line of experiments that are to be conducted."

M. H. Hunt—In regard to the experiments of foundation, I would suggest that there is a great difference in wax, and this alone might account for much of the difference reported.

Pres. Taylor—I know that there is a difference in wax, but I could have all of the foundation made from the same batch. I could make some from it, and then send some to you, some to Dadant, and to others.

Wm. Anderson—There is a great need of experiments in regard to wintering. There is no drawback so great as this, here in the North.

L. A. Aspinwall—I have experimented for 20 years with machinery, and the profits for the last five years have paid for all the experiments. If we could learn how to successfully winter our bees, there would be a saving of thousands of dollars.

Upon motion of Mr. Hunt a committee of three (W. Z. Hutchinson, L. A. Aspinwall and Wm. Anderson) was appointed to draft a resolution expressing the views of the convention, in regard to the desirability of having the experimental work continued, the selection of a man to do the work, and the appointment of a committee to decide in regard to the line of work to be done.

Next came an essay from the Hon. Geo. E. Hilton, on the

Advantages of Northern Michigan for Honey-Production.

That Northern Michigan has advantages over the southern or older portions of the State, none familiar with the productions of honey can deny. But to know the advantages of any locality one must be familiar with the flora. The first advantage to be derived from these newer localities is the early flow. In springs following winters of deep snows our bees are bringing in pollen and some from the willows before the snow is all gone. The soft maples soon follow, then the hard or sugar maple, from which we get large quantities of honey. I have said that I believed were the bees in as good condition to store honey as during the basswood flow, it would come in nearly as fast. The honey very much resembles maple syrup. I think, however, that it gets its color from the mixture of dandelion that comes in at the same time. As I prefer to have this all used in the brood-nest, I do not put on the surplus cases until the raspberry bloom opens, but I have extracted from the stronger colonies' brood-nests to give the queen room, and fed to the weaker ones, and if you have never tried it you would be surprised at the results with the weak colony.

From what I have already written, you will readily see that our bees are in the very best possible condition to store surplus at the opening of the raspberry bloom. The blackberry comes before this is gone, and lasts until clover, and clover lasts until basswood, so you see it gives a continuous flow of white honey from berry bloom to close of basswood.

Some years ago one of the oldest honey-producers in the State (one who lives in the village where they keep the insane and raise celery, and who wintered his bees in a damp cellar, and brought them out in the spring reeking with mold, and declared they wintered splendidly), came to make

me a visit during the berry bloom, before clover was in blossom. We went into the yard, and he remarked that my bees *seemed* strong. "Oh, yes," I said, "they are doing very nicely."

I raised the cover to one of my chaff hives—"What!" he said, "*got surplus cases on?*" "Oh, yes," I said; and stepping to the next hive I showed him one tiered up, and the top one nearly finished.

"Well, if that don't beat me! Why, I had not thought of putting on a super yet," he further remarked. On looking further, he admitted there were more bees in one of my hives than in any three of his.

That year I took 1,500 pounds of berry bloom honey from 65 colonies, but I never knew blackberry to produce so much honey as in that year. My average, that year, was 80 pounds per colony, comb honey.

Two years ago last September, I was sick the entire month. As soon as I could safely get out, I commenced a trip among the bee-keepers, partially for a visit, but principally to buy honey. The most promising fields I found were in Mecosta, Clare, Isabella, Montcalm, Osceola and Lake counties, and the honey I found, as a whole, was of the best quality I ever bought. At Martiney, in Mecosta county, I found a fine lot produced by a young lady. In Clare county I found nice lots, also in Osceola county. In all these counties they get their honey principally from raspberry and willow-herb. The willow-herb coming soon after the berry bloom, and lasting until frost. In all these counties I did not see a section of dark honey, and here comes the sequel to their successful wintering—the brood-chambers are well filled with this white, well-ripened honey, and very little pollen (the willow-herb produces but little pollen); the bees breed up so fast, and the hives are teeming full of good, healthy bees.

In these counties, in the spring and early summer, there are thousands of acres of wild berries, and in the latter part of the summer and fall just as many of the willow-herb.

In Lake county there was less timber, and the golden-rod predominates in the fall, and there is not so much willow-herb. From Baldwin north there are acres and acres of golden-rod, that resemble fields of wheat. If I could be with you, I could tell you much more about this country than I can write,

and it is the first time, I think, in 14 years, that I have missed a State convention.

I wish you all a pleasant and profitable time, which I know you will have.

GEO E. HILTON.

Byron Walker—I have been in the locality of the willow-herb one year, and it did not yield honey that year. I believe it is considered a sure producer of honey. In Clare county there are many asters, and bees have died in the winter. Perhaps the yield was light.

Chas. Koeppen—I believe that more depends upon ventilation than upon the stores. The foul air and moisture must be carried off. I have two apiaries—in one there was a good yield, and in the other but little.

H. L. Hutchinson—I have not had a failure with golden-rod in ten years.

E. G. Grimes—Alsike furnishes the most honey in my locality.

Mr. Koeppen—Alsike is like other plants. Sometimes it furnishes honey, and sometimes not.

H. Webster asked if there was any foundation in the assertion that some bees gathered honey from red clover while others did not.

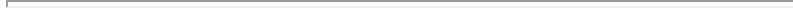
W. Z. Hutchinson—I one year had 1,000 pounds of honey from red clover. It was the result of a drouth that shortened the tubes of the blossoms. I had blacks, hybrids and Italians in the yard, and they all gathered honey from red clover.

August Koeppen said that it would pay to move bees to some other locality only when there was nothing that could be gathered at home. Migratory bee-keeping is largely practiced in Germany.

(Continued next week.)

"**The Honey-Bee:** Its Natural History, Anatomy and Physiology," is the title of the book written by Thos. Wm. Cowan, editor of the *British Bee Journal*. It is bound in cloth, beautifully illustrated, and very interesting. Price, \$1.00,

postpaid; or we club it with the BEE JOURNAL one year for \$1.65. We have only three of these books left.





RANDOM STINGS

FROM "THE STINGER."

The "Stinger's" a poet,
Knows a sheep from a goa-et,
And he stings at random all day;
He thinks he's a honey,
Because he's so funny—
For reference see A. B. J.
—*Progressive Bee-Keeper*

No, I'm not a poet,
Neither did I know-et,
Nor do I sting all the livelong day;
Once a week I've some fun
Making you folks jump and run—
So, what more do you wish me to say?

A certain editor reckoned without his host when he tried to heap more accomplishments upon Editor York than the latter was entitled too. Though Mr. York is a hard working man in the office of the BEE JOURNAL, he is saved the task of doing the stinging; which is, at times, hard work, for some of the people and things that "The Stinger" has to punctuate are pretty tough.

I think the reason why Editor York is not a "Stinger" is because he has not had much to do with the Punics. If he knew from practical demonstrations what those bees were, he would probably become a stinger, too. This is not intended as a joke on somebody's bees.

Rambler was hurt at last. For a time he was confined to the hospital, where I send all those who have been hit with my darts. He announced his injury in the BEE JOURNAL for Dec. 7, 1893, page 730, and he thought the wound inflicted by me must have been produced by a ramrod out of my gun. If he had been hurt as badly as he admitted he was, I am surprised. When my sting penetrated his thick hide he must have seen stars, consequently, at the same time, he had no trouble in magnifying a sting into the proportions of a ramrod. I would say in a fatherly way: My dear Rambler, keep your nose from rambling around in the loose way that you have been letting it stray about, and there will be little danger of its running up against the sting of The Stinger.

Rambler says he smiled a "smole" when his nose came in contact with my "stinger." Stings seem to have the same effect upon him that laughing-gas has upon a patient in a dentist's chair. Rambler, beware, for have you not heard that "laughing often comes to crying?" The next time we may hear from you, you may be sitting in a corner crying, because the sting got into

your nose a little below the tip, and it is hurting you in a way that a sting never troubled you before.

The mission of The Stinger is to reform the bee-keeping world. (Did I hear you say that that is impossible, dear reader?) There is no man in the world that needs reforming more than the Rambler, and so his threats to expose me if I do not cease troubling him, fall upon me as uselessly as if he had never uttered them. For shame, on you, Rambler! to intimate that you will silence my pen. You might as well try to melt the snow on the tops of the high mountains away back of where you live, with that genial smile of yours, as to keep The Stinger from performing his mission. Rambler, beware of the day when I shall meet you in battle array.

Some one has sent me a copy of the December number of the *California Cultivator and Poultry Keeper*. It is a nice publication, but I do not see how it manages to live under such a load of a name. As it has a well edited apiary department, I imagine that the publisher will some day add *Bee-Keeper* to the already long title.

A correspondent writes to know if The Stinger is a woman; she says she thinks The Stinger must be a female, because males do not sting. I would inform the fair writer, and all other persons who have doubts as to the sex of The Stinger, that he is a male; this male stings, if other males do not.

What is the difference between a swarm of bees and a sewing bee?

None, as far as buzzing is concerned.—*Ex.*

Charlie—"Papa, why is it that honey, money and funny rhyme?"

"I don't know, unless that it is often very funny to get honey out of a beehive when the bees are all about your head, and because it is worth all the money one gets for it to get the honey from the bees. I heard a man say that he would not take honey from bees at any price."

Charlie—"Well, that's funny."

Student in apicultural class at agricultural college—"Professor, why is it dangerous for a person with the blues to go into an apiary?"

Professor (perplexed)—"I do not know; the text-books do not say anything on the subject."

Student (with much glee)—"Because bees are said to have a preference for blue!"

"To be or not to be stung," might have been written by Shakespeare instead of all that stuff about shuffling off this mortal coil, that school boys are so fond of spouting on declamation days. If he had said that about the bees, and a little more too, we might now be classing the Bard of Avon as something of a bee-keeper; and perhaps we would be having a peep into his immortal works through the pages of our friend, *Gleanings*.

Mr. Maybee—I have read that a professor in one of our agricultural colleges says that there is considerable difference between the sting of a wasp and that of a bee.

Mrs. M.—I am not willing to take that learned man's word for it, as I was stung by both, and did not see the difference; both are too hot for me.



LANGSTROTH FUND.



[For years, bee-keepers have felt that they owed the Rev. L. L. Langstroth—the Father of American bee-culture—a debt that they can never very well pay, for his invention of the Movable-Frame Hive which so completely revolutionized bee-keeping throughout all the world. In order that his few remaining years may be made as happy and as comfortable as possible, we feel that we should undertake a plan by which those bee-keepers who consider it a privilege as well as a duty, might have an opportunity to contribute something toward a fund that should be gathered and forwarded to Father Langstroth as a slight token of their appreciation, and regard felt for him by bee-keepers everywhere. No amount above \$1.00 is expected from any person at one time—but any sum, however large or small, we will of course receive and turn over to Father L. All receipts will be acknowledged here.—ED.]



List of Contributors.

Previously Reported	\$79 95
Margaret Swain, Pendleton, Ind.	50
Ed. Weidner, Earlville, Ills.	1 00
Scott LaMont, Jarrett, Minn.	90
Wm. Kittinger, Caledonia, Wis.	1 00
Total	\$83 35

CLUBBING LIST.

We Club the *American Bee Journal* for a year, with any of the following papers at the club prices quoted in the **LAST** column. The regular price of both is given in the first column. One year's subscription for the American Bee Journal must be sent with each order for another paper:

	<i>Price of both.</i>	<i>Club.</i>
The American Bee Journal	\$1 00	
and Gleanings in Bee-Culture	2 00	1 75
Bee-Keepers' Review	2 00	1 75
Canadian Bee Journal	2 00	1 75
The Apiculturist	1 75	1 65
Progressive Bee-Keeper	1 50	1 30
American Bee-Keeper	1 50	1 40
Nebraska Bee-Keeper	1 50	1 35
The 8 above-named papers	6 25	5 25

Have You Read page 101 yet?



OUR LETTER BOX

REPORTS PROSPECTS &c



Do not write anything for publication on the same sheet of paper with business matters, unless it can be torn apart without interfering with either part of the letter.



The "Bee Journal" a Great Help.

There is so much valuable information in the BEE JOURNAL that I cannot afford to destroy them, but will bind them and keep them as a book of reference. Really, I don't see how I could do without the BEE JOURNAL, or some other bee-paper that would come up to its standard. Last year I produced 1,500 pounds of comb honey, while in former years I never got over 400 pounds. Of course the extra flow last year accounts for much of my better success, but I also attribute very much of it to the "Old Reliable."

Myself and son have now 55 colonies in winter quarters, all apparently doing well except one colony that I think is queenless. Our success last year would have been much greater had the dry weather not cut the basswood flow short.

A. H. SNOWBERGER.

Huntington, Ind., Jan. 5, 1894.

Had Plenty of Good Flights.

I had 5 colonies, spring count, and increased to 11. They did very well the forepart of the season, but it was too dry to grow buckwheat, or any other honey-plant. The bees went into winter quarters in good condition, and are all right up to date. They have had plenty of good flights so far, but the worst is to come yet. In the spring, if the weather stays warm, they consume more than when it is cold. Bees were rather scarce last spring.

W. F. RINCK.

West Alexandria, O., Jan. 1, 1894.

A Good Report—Bee Management.

My 18 colonies of bees came through the winter in good condition last spring. They gave me a surplus of 2,000 pounds of white and sweet clover honey, 1,400 pounds of extracted, and 600 pounds of comb honey. The former sold here at \$1.00 a gallon, and the latter at 13 cents a pound. I run 5 colonies a different way for extracted honey, and those 5 gave a surplus of 800 pounds. It was done as follows:

I watch until they prepare to swarm, and the honey-flow is close. I take out all frames from the brood-chamber, except the one the queen is on, which I put in the center, and fill the chamber with new frames of full sheets of comb foundation. I then take a full sheet of Root's perforated zinc, with $\frac{1}{4}$ -inch bee-space between the frames and zinc, and put it over the brood-chamber. I then put a chamber on top of the zinc, and put the frames with the bees and brood in this top chamber, and cover it up. Now I have a laying queen and lots of room for brood below, and as fast as the brood hatches

above, they fill it with honey if the flow is here. It was here this year, for they filled the top chamber, after the first extracting, in four days—6 frames two-thirds capped.

My increase is from 18 to 25 colonies, which are in double-walled hives, and in as good condition for winter as I ever had them.

HENRY BOHLMANN.

Defiance, O., Jan. 1, 1894.

Uses of Perforated Zinc.

In reading the short item by Mrs. Jennie Atchley, in regard to the different uses of Dr. Tinker's perforated zinc, I thought I would add a little of my experience to those already given.

1st. In hiving young swarms, I have found it to be excellent to place over the entrance to keep them from leaving or returning to the home hive.

2nd. When four or six swarms issue at the same time, and cluster together, I have found it to be of the greatest value to me. I look the bees over, find my queens, and place them in separate hives, and put on the zinc over the entrance; then I take a large dipper and dip the bees from the place where they alighted, putting them in front of the different hives, when the bees will separate, each swarm going into its own hive.

In using the zinc, some might misunderstand me. I only leave the zinc on the entrance from two to four days; if the queens are young, I only leave it on two days, so as to give them a flight. With old queens I leave it on longer.

ANDREW M. THOMPSON.

Canaseraga, N. Y.

Something from Central California.

Being a Californian, and having not as yet crossed the State line, it is with great pleasure that I read the reports from other States or localities

throughout the United States. I was pleased to see in the BEE JOURNAL a report from Kern county—a county joining Tulare county on the south—of my neighbor realizing nearly 300 pounds per colony. Now I have reasons to believe that bees will produce a great deal of honey per colony in Kern county, as I own a small ranch containing 685 acres down there, and am quite familiar with the country. We have the largest alfalfa fields in the world in Kern county, Calif. We have a man in Kern county who owns in one body almost 1,000,000 acres of land.

I have seen the time that all the counties in the San Joaquin valley, consisting of Kern, Tulare, Kings, Fresno, Modara, Merced, and Modesto—all were joined in large tracts, and had their herds after herds of bellowing cattle roaming over its one level plain, as the San Joaquin valley is level, not one elevation 50 feet high in a valley that is 75×200 miles in size.

Our Senator, Tom Fowler, who owned cattle all along the coast from San Francisco to Los Angeles, used to say: "I own the cattle that roam on a thousand hills." I am the same old 76. Tom and all of his bellowing herds are no more. The "76 ranch," which is located in Tulare county, was Tom's head-quarters. It has been cut up into small farms, all the way from 20 up to 2,000 acres, and there are thousands of happy and beautiful homes, school houses, churches and towns, instead of the mustang and its master.

Our part of the State is not generally known, as this is central California, and the cities north and south try to claim us as theirs.

FRED M. HART.

Traver, Calif.

A Little Experience with Bees.

Bee-keeping has about "gone to seed" in this part of the country. If a person undertakes to keep bees on modern principles, he is considered a crank. They think "pa" knew it all. "Pa" kept 15 or 20 colonies, and would get 20 or 25 pounds of honey from one colony. "Pa" would take care of his bees; he would put little blocks or nails under each corner of the hive, and then moths would not bother them!

I have read Quinby's book, "A B C of Bee-Culture," *Gleanings* for two years, the BEE JOURNAL for one year, also "Bees and Honey," and with my practical experience, I feel that I am just beginning to learn my A B C's.

My experience is not very extensive. I ran one colony this year for increase, and made 10 colonies from it, and they only cast one natural swarm. I got the idea of artificial increase from "Bees and Honey." That alone has been worth ten times the price of the BEE JOURNAL to me. Those 10 colonies of bees could not be bought for \$20. People will say "times are too hard—I can't spare the money."

I can't close without telling how much I enjoy "In Sunny Southland." That alone is worth the price of the paper. Long live the AMERICAN BEE JOURNAL!

C. L. DOYLE.

Fayette Corners, Tenn.

Half a Crop—Golden Italians, Etc.

The common verdict regarding the honey season in this locality is but half a crop. A cold, wet and late spring left the colonies in such a weak condition, that were they not stimulated, they would have been in no condition at all when the honey-flow came. We had a splendid flow from honey-locust, although there were such high winds and continual rains during the bloom, that the bees could work scarcely a day at a time.

Right here I want to say one good thing for those golden 5-banded Italians, which race almost every one wishes to condemn. They were working almost every day, while the others did not dare venture out. This shows them to be very strong on the wing, but as to their superiority as everyday honey-gatherers, over the leather-colored variety, I am not prepared to say. For a hardy strain, long livers, and a business class of bees, give me uniformly marked leather-colored Italians, every time.

To return to our honey-resources: White clover was almost a total failure, as a severe drouth existed during the bloom, and it did not secrete much nectar. Smartweed was our main stand-by, with golden-rod a close second. They yield an excellent honey.

My style of marketing is three one-pound sections in a frame, for which I had no trouble in obtaining 60 cents. Extracted brings 12½ cents, although there is complaint of adulteration in our market.

Our bees are in fine condition for winter, and we have hopes for a better season next year.

J. C. WALLENMEYER.

Evansville, Ind., Dec. 17, 1893.

Good Season in 1893, Etc.

A queer winter we are having thus far. The forepart of December was cold, the mercury reaching zero a number of times, with very little snow. The middle of December it warmed up, the snow all disappeared, and on the 22nd it was 70 degrees in the shade; on the 25th it was 60 degrees, and I let my bees have a flight. They were not as thick as in June.

The last season was a good one in this locality. I never saw white clover so thick before. We had a heavy wind and hail storm in buckwheat bloom, which was a complete stop to the buckwheat flow, which started in well. Golden-rod and asters did not yield much. There were a good many runaway swarms the past season, quite a number being found on the lake shore. I got four. The lake takes off one-half, or nearly that, of my pasture (being situated on the shore). We have about 8 inches of ice now, and have had very good ice-boating so far.

S. H. EASTWOOD.

Cicero Centre, N. Y., Jan. 8, 1894.

Won't Winter on Sorghum.

On page 559, of the BEE JOURNAL for 1893, there is an item concerning sorghum for wintering bees. Mrs. Atchley suggests that I try it and report, which I will do with pleasure.

I can only report failure. Mrs. Atchley reports that she could not get her bees to take hold of sorghum. I had 5 colonies which I fed on it last

October, sufficient to carry them through the winter. They are now all dead but one colony, and that one is reduced in numbers to a mere handful of bees. They all had plenty of sealed stores when they died, and fell down on the bottom-board. All of my other colonies are wintering well, that have honey stores.

Now, if Mrs. Atchley wants to try sorghum next winter, I think if she will go to some of her colonies of bees in warm, dry weather in October—some that have plenty of bees and not much honey—and raise the front end of the hive a little higher than the back, and pour in the pure sorghum just a little for one or two evenings, to get them started to eating it, then increase a little more, feeding every evening, I think in a week or so her bees will have plenty of sealed stores to last them through the winter—if they should live that long. But I don't think they would.

I don't wish Mrs. Atchley to feed her bees on sorghum, nor any one else, unless you want to lose your bees, for that you will do if they are fed on pure sorghum.

Some Northern bee-keepers may think there is a disease among my bees, but such is not the case. There never was any disease among bees in this country, that I know of.

N. E. CLEVELAND.

Decatur, Miss., Dec. 23, 1893.

Getting Statistics on Bee-Culture.

I notice on page 743 of the BEE JOURNAL for 1893, under heading of "Comb Honey in the United States," a request for all manufacturers of honey-sections to report all sales of sections to Dr. Miller, for the purpose of ascertaining the amount of honey produced, etc. That would be one way to guess at the amount, but I don't think it would be very much of a guess.

It appears to me that there is but one way to get at the amount of honey produced in the United States. Every assessor has a long list of questions provided on purpose to get at the statistics of the country. When these statistics are finally compiled, they are sent out all over the country, and we can see at a glance just how much wheat, oats, corn, etc., each State has

produced the previous season. Now, I don't think it would require very much persuasion on the part of the bee-fraternity to secure the placing of two or three more questions on that list, viz.:

1. How many colonies of bees did you have, spring count, on June 1st, last year?
2. How many pounds of comb honey did you produce?
3. How many pounds of extracted honey?

This would bring out a full report of the amount of honey produced in the United States. It would also show the number of colonies of bees kept by the States.

C. H. POND.

Kasson, Minn.



Value of Bee Papers and Books.

There is little use trying to keep bees, either for pleasure or profit, without at least one live bee-paper to awaken interest and enthusiasm, and keep the apiarist abreast of the times. When we see an apiary that shows neatness, taste and prosperity, we need not be told that the owner or manager has access to bee papers and books; and when we find a bee-yard with hives huddled together regardless of order, distance or taste, with many of the colonies dead and dying, we are sure that the knowledge, skill and enterprise that come from the study of apiarian literature, have never reached that desolate and forlorn spot. Of course the bees are black, but no darker than much of the filthy comb and honey inside the hives. By neglect, much of the worker comb has become unfit for brood-rearing, and hence drones are reared in superabundance, and these deteriorated male bees fill the air for miles in all directions, to vitiate the pure blood of all well-kept apiaries.

But the intelligent, careful, painstaking apiarist will find encouragement in the assurance that all bees kept by such slipshod methods are doomed, and on the principle of "the survival of the fittest" must go, and the sooner the better for all concerned.

Bees in this section did well last season, and went into winter quarters in excellent condition. The recent warm spell gave them a fine airing (those on the summer stands), and now they should winter with but little loss.

S. S. BUTTS.

Wyalusing, Pa., Dec. 28, 1893.

The Season of 1893, Etc.

I read the BEE JOURNAL with much pleasure, for my bees are almost all the comfort that I have left, as I have no family now. My wife died last February. I have two daughters, but they were married years ago. I have a comfortable home, and enough of this world's goods to live on, but what matters that, when the ties of love are forever broken?

Well, the last season was not very good for the bee-man in this part of the State. The dry weather set in just as the basswood came into bloom, and cut it short. It was just a little cool for white clover, so our crop was short. I have always worked for comb honey, and for that reason I have never had very heavy returns.

My bees came out of the cellar in fair condition last spring. I put away 20 colonies, and lost 4 through my neglect (the breaking up of my family unnerved me for business). The bees increased to 30 colonies, which are now in the cellar, as that is my method of keeping them. They are heavy with winter stores, and so far are doing well. They were put in on Nov. 22nd.

I want to say a few words in favor of the yellow bee, as I have both. They stored nearly all the surplus. In a good season the blacks will do just as well, but when the crop is short, the yellow bees are the best for me.

I have sold 500 pounds of honey, and have 100 or more of uncapped honey left.

D. C. WILSON.

Viola, Iowa, Dec. 19, 1893.

Great Premium on page 101!

TRANSCRIBER'S NOTES.

1. Silently corrected simple spelling, grammar, and typographical errors.
2. Retained anachronistic and non-standard spellings as printed.

*** END OF THE PROJECT GUTENBERG EBOOK THE AMERICAN
BEE JOURNAL, VOLUME XXXIII, NO. 4, JANUARY 25, 1894 ***

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